Code

5.1

```
clc; clear; close all;

%% --- Input Parameters ---

fp = 1000; % High-pass cutoff frequency (Hz)

fs = 8000; % Sampling frequency (Hz)

N = 50; % Filter order


wc = fp/(fs/2); % Normalized cutoff (0–1)


%% --- 1. FIR High Pass using Rectangular Window ---

b_rect = fir1(N, wc, 'high', rectwin(N+1));

disp('--- High-Pass Coefficients (Rectangular Window) ---');

disp(b_rect');


%% --- 2. FIR High Pass using Hamming Window ---

b_hamm = fir1(N, wc, 'high', hamming(N+1));

disp('--- High-Pass Coefficients (Hamming Window) ---');

disp(b_hamm');

%% --- 3. FIR High Pass using Hanning Window ---

b_hann = fir1(N, wc, 'high', hanning(N+1));

disp('--- High-Pass Coefficients (Hanning Window) ---');

disp(b_hann');

%% --- Frequency Response Plots ---

figure;

freqz(b_rect,1);

title('High-pass FIR using Rectangular Window');

figure;

freqz(b_hamm,1);

title('High-pass FIR using Hamming Window');
```

```
figure;
freqz(b_hann,1);
title('High-pass FIR using Hanning Window');
```

Output

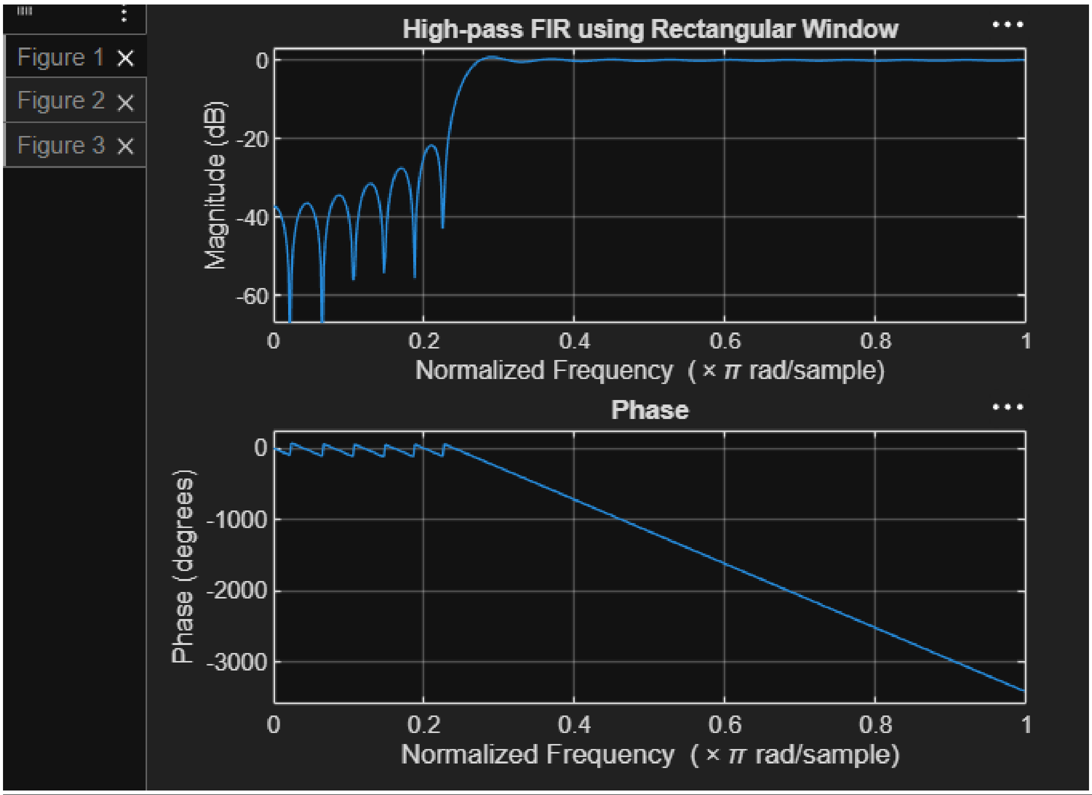

Figure 1
Figure 2
Figure 3
High-pass FIR using Rectangular Window
Magnitude (dB)
0
-20
-40
-60
0
0.2
0.4
0.6
0.8
1
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
0
-1000
-2000
-3000
0
0.2
0.4
0.6
0.8
1
Normalized Frequency (× π rad/sample)

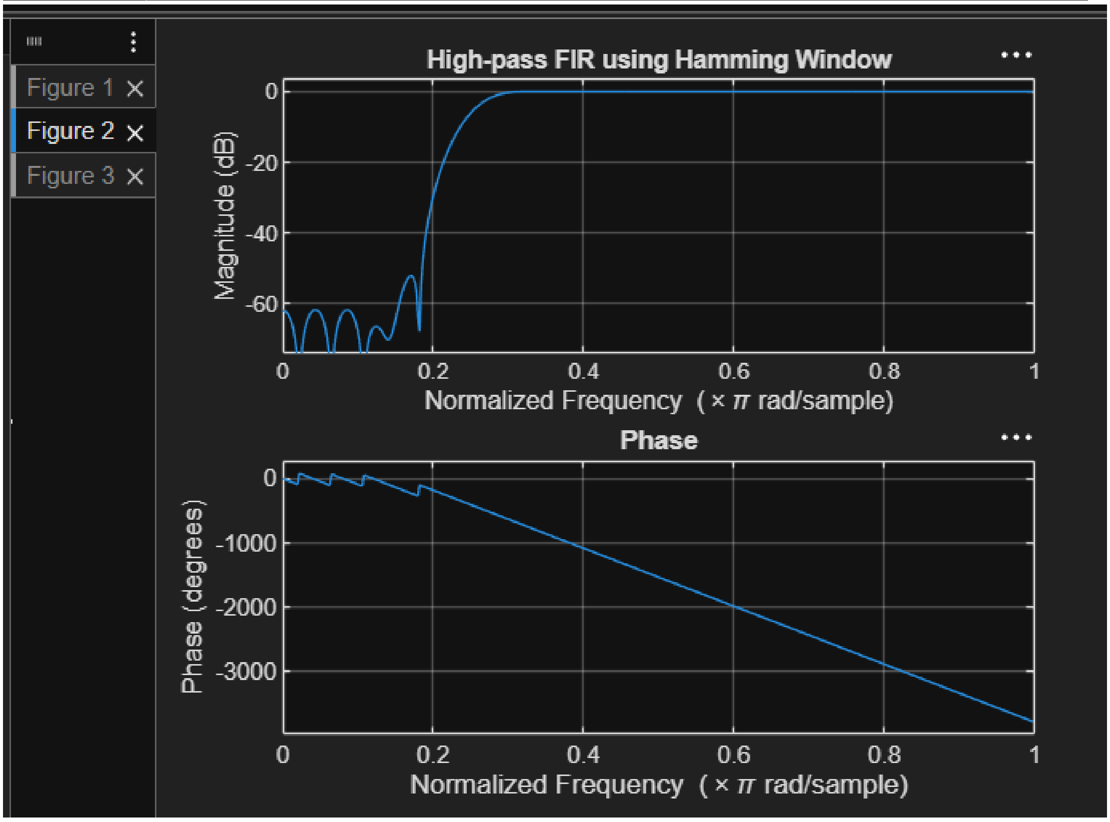

Figure 1
Figure 2
Figure 3
High-pass FIR using Hamming Window
Magnitude (dB)
0
-20
-40
-60
0
0.2
0.4
0.6
0.8
1
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
0
-1000
-2000
-3000
0
0.2
0.4
0.6
0.8
1
Normalized Frequency (× π rad/sample)

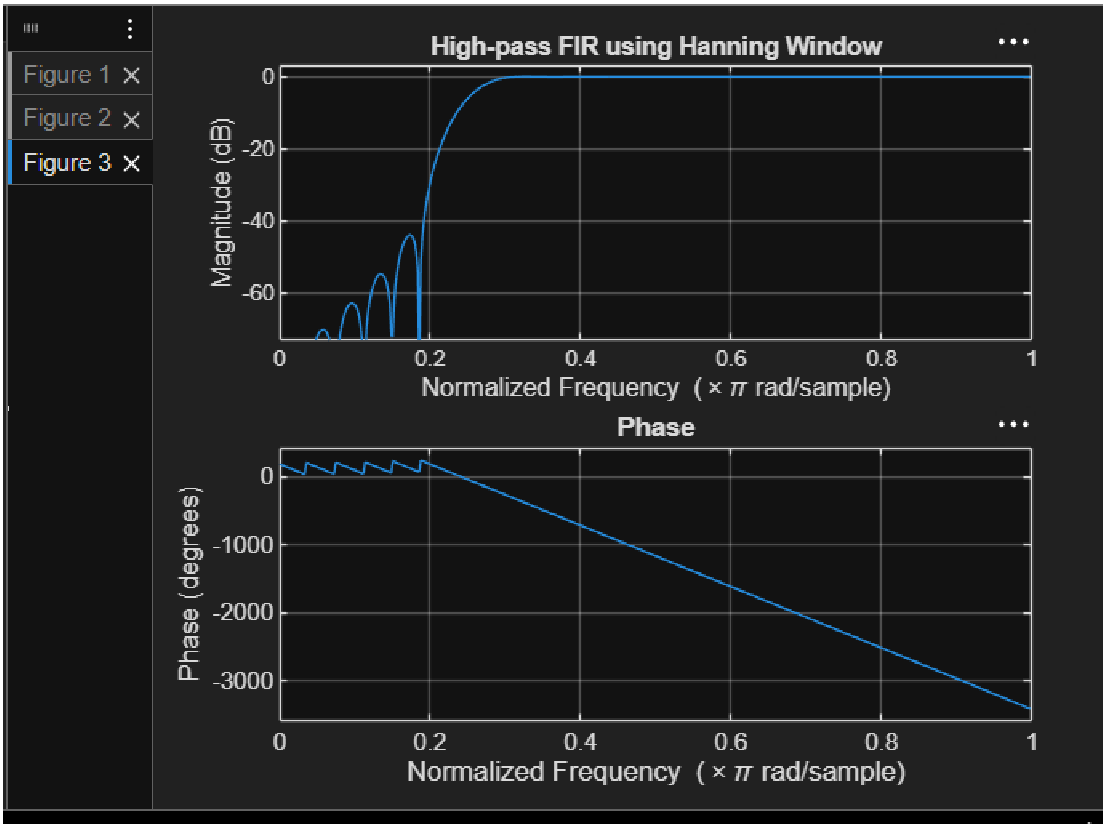
Figure 1
Figure 2
Figure 3
High-pass FIR using Hanning Window
Magnitude (dB)
0
-20
-40
-60
0
0.2
0.4
0.6
0.8
1
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
0
-1000
-2000
-3000
0
0.2
0.4
0.6
0.8
1
Normalized Frequency (× π rad/sample)